

Angular Curriculum



COURSE CURRICULUM

Angular Framework Documentation

1. TypeScript Introduction

- Overview of TypeScript
- Why TypeScript for Angular
- TypeScript vs JavaScript
- Setting up TypeScript Environment

2. Key Concepts in TypeScript

3. Angular Installation

- Prerequisites (Node.js, npm)
- Installing Angular CLI
- Creating a New Angular Project
- Running and Building the Application

4. Understanding Angular Project Structure

- Overview of Angular Files and Folders
- main.ts and app.module.ts
- Components, Modules, and Templates
- Best Practices for Folder Organization

5. Routing and Advanced Routing

- Introduction to Angular Routing
- Configuring Routes in app-routing.module.ts
- Route Parameters
- Route Guards and Navigation Control

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6. Data Binding and Its Types

- Concept of Data Binding
- Interpolation – Displaying dynamic data
- Property Binding – Binding DOM properties
- Event Binding – Listening to user actions
- Two-Way Binding – Synchronizing data

7. Directives and Their Types

- What are Directives
- Structural Directives – *ngIf, *ngFor, *ngSwitch
- Attribute Directives – ngClass, ngStyle
- Creating Custom Directives

8. Pipes and Their Types

- Introduction to Pipes
- Built-in Pipes – date, uppercase, currency, etc.
- Custom Pipes – Creating and Registering
- Async Pipe for Observables

9. Forms in Angular

- Forms Overview and Differences
- Template-Driven Forms – Simplicity and Syntax
- Reactive Forms – FormBuilder and FormGroup
- Form Validation (Built-in & Custom)
- CRUD Operation Using Reactive Forms and Template Forms
- Custom Toast Service

10. Decorators in Angular

11. Services in Angular

- Concept of Services and Single Responsibility
- Creating and Registering a Service
- Using Services Across Components

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12. HTTP Client and API Integration

- Introduction to Angular HttpClient
- Performing GET, POST, PUT, DELETE Requests
- Handling Observables and RxJS Operators
- Managing API Errors and Responses
- Integrating with RESTful JSON APIs

13. Authentication with Local Storage

- Overview of Client-Side Authentication
- Login and Logout Functionality
- Managing User Sessions
- Handling Unauthorized Access

14. Authorization and Role-Based Guards

- Introduction to Angular Guards
- Implementing CanActivate and CanActivateChild
- Creating Role-Based Access Controls
- Protecting Routes Based on Roles

15. Student Management Project (CRUD Example)

- Project Setup and Overview
- Creating Components and Models
- Implementing CRUD (Create, Read, Update, Delete) Operations
- Integrating Reactive Forms
- Applying Role-Based Guards
- Final Project Workflow



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